

Benjamin Curis-Friedman

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EDUCATION

McGill University 3.96/4.0 GPA

Montreal, QC

Bachelor of Software Engineering – Minor of Mathematics

Aug. 2024 – May 2028

Awards (\$6,000): Alma Mater Entrance Scholarship, Class of Engineering '83 Scholarship

EXPERIENCE

Software Developer Intern

Jan. 2026 – Aug. 2026

Rogue Research Inc

Montreal, QC

- Contributed to Pyvista & VTK, improving rendering performance over **50x**, and bug fixes.
- Implemented **GLSL vertex shaders** for realistic optical distortion in real-time 3D scene rendering.
- Engineered a **3D surgical robot emulator**, enabling accurate visualization and simulation for surgery planning.
- Refactored a robot's **networking packet parser** to safely handle bad data, improving communication reliability.
- Reimplemented 2 robotic brain-stimulation research papers using **Bayesian optimization and BoTorch**.
- Implemented the **SSIM** image comparison algorithm and improved unit-test coverage by **~20%**.

AI Researcher Intern

May 2025 – Dec. 2025

iSMART Lab

Montreal, QC

- Developed a python video synchronization & compression library using FFmpeg and OpenCV, reducing raw dataset storage size by over **60%** while keeping frame-level sync accurate to **<10ms**.
- Coded a **data pre-processing** pipeline in Python and Scipy to clean raw human vital signals with band-pass filters, increasing the signal-noise ratio and cutting manual review time for the research team.
- Built and programmed an embedded task labeling device using Arduino and C++, cutting recording segmentation time from hours to minutes per session across **50+ hours** of multi-modal study data.

CERTIFICATIONS

AWS Certified Solutions Architect - Associate (SAA-C03)

Sept. 2026

Machine Learning Specialization - Stanford Online

Jul. 2025

PROJECTS

Omniclave | *FastMCP, FastAPI, Auth0, React, Bun, Codex*

May 2026

- Built an **MCP** server for a student portal, harnessing agents to help with homework and emails.
- Integrated **OAuth2.0** to let users login from their favourite MCP client (e.g. Codex / Claude code).
- Automated a seamless login flow using a backend playwright instance to log into Omnivox on a user's behalf.

CheeseOps | *Java, JavaFX*

Sept. 2025

- Developed an application for a comté cheese distribution business following the **MVC pattern**.
- Designed a domain model through **UML class diagram** and state machine using Umple.
- Created reusable button, menu, and modal components in JavaFX.

AI Library From Scratch | *Python, NumPy, Keras, PyGame*

Jul. 2025

- Coded a neural network library complete with mini-batch **gradient descent** and activation/cost functions.
- Trained a sequential neural network achieving **96% test accuracy** on MNIST dataset.
- Implemented a **modular network architecture**, allowing custom network shapes like TensorFlow.

WebChat | *Firebase, Javascript, Vue.js, HTML, CSS*

Sept. 2023

- Developed a full-stack web messaging platform with authentication for **70+ users**.
- Programmed search and add friends features with **Google Cloud Functions and Firestore DB**.
- Implemented Firestore security rules to ensure secure authenticated & authorized database access.

SKILLS

Languages: Python, OCaml, C, Objective-C, Java, SQL, Bash, C++, TypeScript, JavaScript, HTML, CSS, R

Frameworks/Cloud: AWS, Google Firebase, React, Node.js, JUnit, Tailwind, Cocoa, JavaFX

Developer Tools: Git, Github Actions (CI/CD), Docker, CDash, VS Code, XCode, NeoVim, Linux, Subversion

Libraries: NumPy, Matplotlib, PyVista, VTK, OpenGL, OpenCV, pandas, Ax Platforms, SciPy, Keras, TensorFlow

Spoken Languages: French, English